MIL-C-13777/3A
1 September 1966
SUPERSEDING
MIL-C-13777/3
25 March 1963

MILITARY SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL 16, 17, 18, 19, AND 20 CONDUCTORS

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-13777.

REQUIREMENTS:

Dimensions and configuration: See applicable figure and design data for the following cable types:

160636	S1613758	180675	S1909658	200935
S1608788	170752	180848	1911108	
1610658	170874	S190875	191140	

FSC 6145

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DESIGN DATA

Type Designation	160636			S160878S			161065S	
Figure No.	1			2			3	
Total Wires	16			16			16	
No. of Conductors & AWG #	16/#20	8/#18			8/#18S	12/#18		4/#18*
Insulation								
Min average thickness	0.012"			0.015"			0.015"	
Spark Test Voltage	3000			3000			3000	
Inspection Test Voltage	1500			1500			1500	
Coax. Jacket Insulation								
(a) Min average thk.							0.025"	
Cabling								
Layer No. 1	Filler			Filler			Filler	
Layer No. 2								
(a) Number of wires	5			6			4	
(b) AWG #	#20			#18			#18S	
(c) Maximum Lay	2.0"			2.50"			3.75"	
Layer No. 3								
(a) Number of wires	11	2			8		12	
(b) AWG #	#20	#18			#18S		#18	
(c) Maximum Lay	4.5"			5.00"			6.00"	
Sheath								
No. of Layers	2			1			2	
Total thickness Min	0.109"			0.080"			0.100"	
Minimum OD Cable	0.616"			0.863"			1.050"	
Maximum OD Cable	0.656"			0.893"			1.080"	

Note: * Jacketed Coaxial Wires

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DESIGN DATA

Type Designation		S1613758		170752	170874
Figure No.		4		5	5
Total Wires		16		17	17
No. of Conductors & AWG #	2/#18S	10/#18S	4/#6	17/#18	17/#16
Insulation					
Min average thickness		0.015"	0.060"	0.015"	0.015"
Spark Test Voltage		3000	7500	3000	3000
Inspection Test Voltage		1500	2500	1500	1500
Cabling					
Layer No. 1				Filler	Filler
(a) Number of wires		2			
(b) AWG #		#18*			
(c) Maximum Lay		3.00"			
Layer No. 2					
(a) Number of wires	5	5	4	6	6
(b) AWG #	#18**	#18S	#6	#18	#16
(c) Maximum Lay		8.00"			
Layer No. 3					
(a) Number of wires				(1)	(2)
(b) AWG #				#18	#16
(c) Maximum Lay				5.00"	6.00"
Sheath					
No. of Layers		2		2	2
Total thickness Min		0.100"		0.110"	0.109"
Minimum OD Cable		1.360"		0.737"	0.849"
Maximum OD Cable		1.390"		0.767"	0.899"

Note: *Shield applied over twisted pair of AWG #18 -- Max Twist Lay 3.00"

**Assembly with max lay of 3.00"

(1) 11 & 1 Filler

(2) 11 W/Fillers

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DESIGN DATA

Type Designation	180675		180848		S190875
Figure No.	6		7		8
Total Wires	18		18		19
No. of Conductors & AWG #	18/#20	14/#18		4/#16	19/#18
Insulation					
Min average thickness	0.015"		0.015"		0.015"
Spark Test Voltage	3000		3000		3000
Inspection Test Voltage	1500		1500		1500
Cabling					
Layer No. 1	Filler		Filler		Filler
Layer No. 2					
(a) Number of wires	6	2		4	7
(b) AWG #	#20	#18		#16	#18
(c) Maximum Lay					
Layer No. 3					
(a) Number of wires	12		12		(1)
(b) AWG #	#20		#18		#18
(c) Maximum Lay					5.00"
Sheath					
No. of Layers	1		2		1
Total thickness Min	0.080"		0.098"		0.094"
Minimum OD Cable	0.650"		0.823"		0.850"
Maximum OD Cable	0.700"		0.873"		0.900"

(1) 12 W/Filler

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DESIGN DATA

Type Designation	S1909658				1911108		
Figure No.	9				8		
Total Wires	19				19		
No. of Conductors & AWG #	8/#18S		11/#18	7/#20		12/#20S	
Insulation							
Min average thickness	0.015"				0.012"		
Spark Test Voltage	3000				3000		
Inspection Test Voltage	1500				1500		
Cabling							
Layer No. 1	Filler				Filler		
Layer No. 2							
(a) Number of wires	8				7		
(b) AWG #	#18				#20		
(c) Maximum Lay	2.50"				4.0"		
Layer No. 3							
(a) Number of wires	8		3		12		
(b) AWG #	#18S		#18		#20S		
(c) Maximum Lay	5.00"				8.0"		
Sheath							
No. of Layers	1				2		
Total thickness Min	0.080"				0.140"		
Minimum OD Cable	0.950"				1.080"		
Maximum OD Cable	0.980"				1.140"		

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	DESIGN DATA		
Type Designation	191140	200935	
Figure No.	8	10	
Total Wires	19	20	
No. of Conductors & AWG #	19/#12	13/#18	7/#15
Insulation			
Min average thickness	0.020"	0.015"	
Spark Test Voltage	4000	3000	
Inspection Test Voltage	2000	1500	
Cabling			
Layer No. 1	Filler	Filler	
Layer No. 2			
(a) Number of Wires	7	7	
(b) AWG #	#12	#15	
(c) Maximum Lay		3.25"	
Layer No. 3			
(a) Number of wires	12	13	
(b) AWG #	#12	#18	
(c) Maximum Lay		4.25"	
Sheath			
No. of Layers	2	2	
Total thickness Min	0.098"	0.125"	
Minimum OD Cable	1.115"	0.920"	
Maximum OD Cable	1.165"	0.950"	

Custodians:
Army - MU
Navy - SH
Air Force - 17

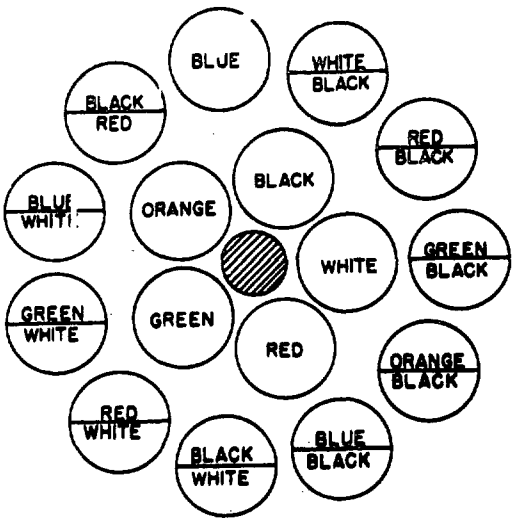
Preparing activity:
Army - MU

(Project 6145-0420)

Reviewer:
Army - MI, EL, MO
Navy - SH
Air Force - 15

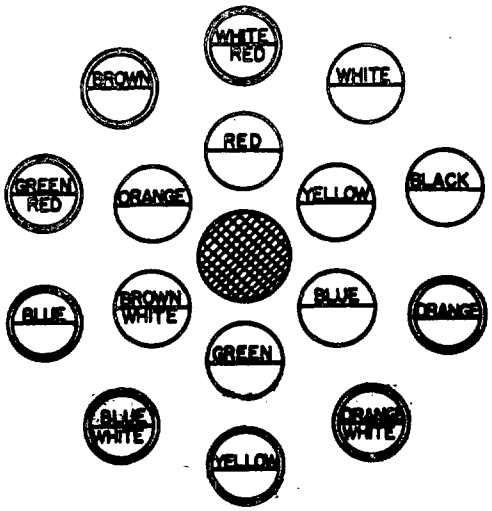
Users:
Army - CE
Navy - MC
Air Force - 11

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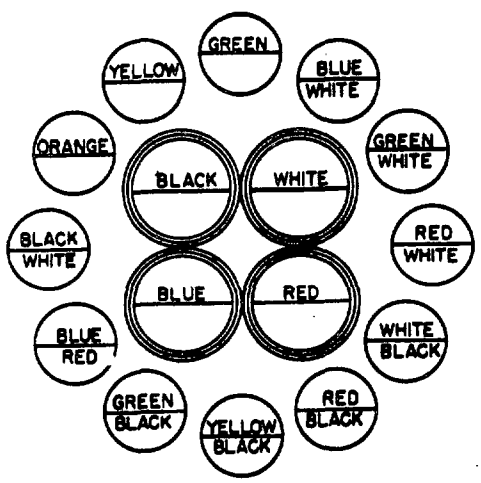
CABLE TYPE: 160636

FIGURE 1



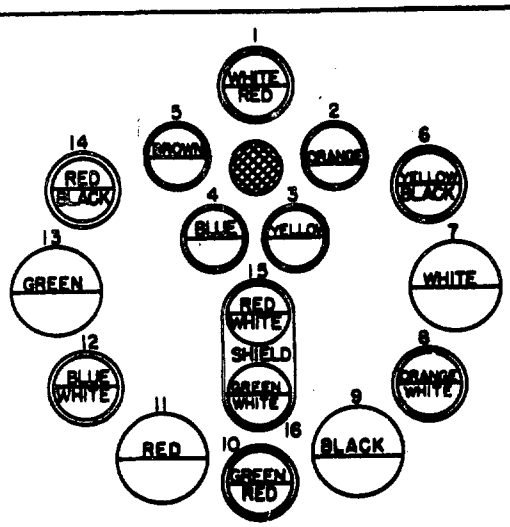
CABLE TYPE: S160878S
NOTE: DOUBLE CIRCLE INDICATES SHIELDED CONDUCTORS

FIGURE 2



CABLE TYPE: 161065S
NOTE: TRIPLE CIRCLE INDICATES COAXIALS WITH DRIVEN SHIELDS

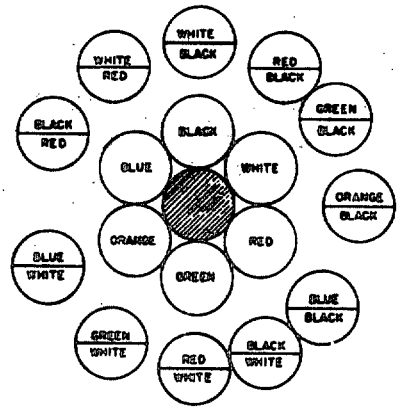
FIGURE 3



CABLE TYPE: S161375S

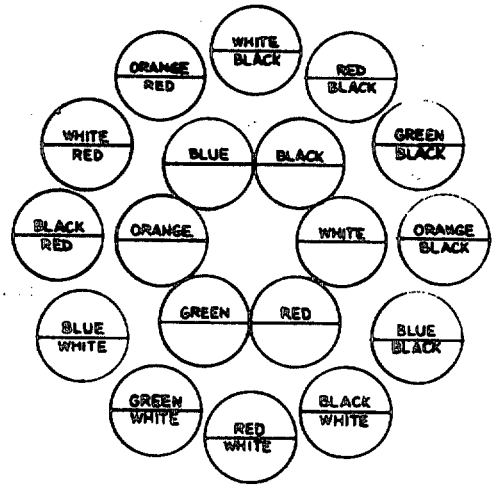
FIGURE 4

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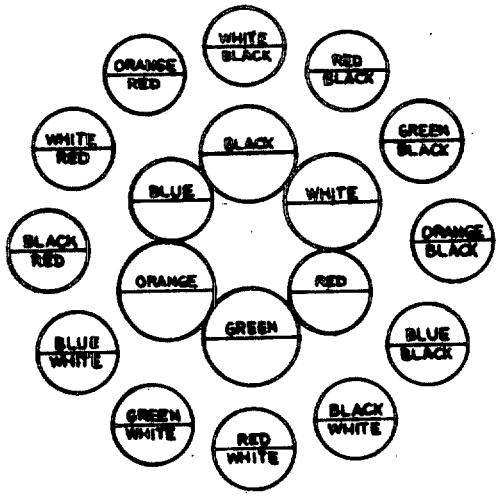
CABLE TYPES: 170752 170874

FIGURE 5



CABLE TYPE: 180675

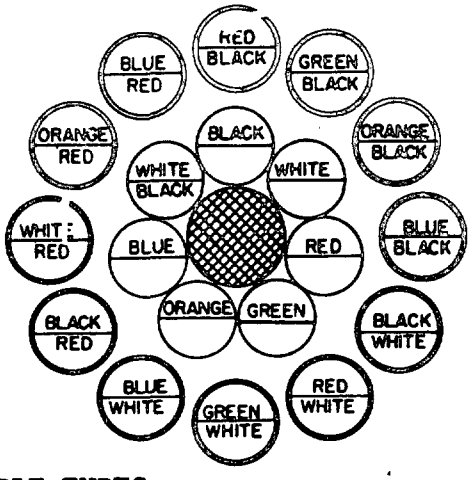
FIGURE 6



CABLE TYPE: 180848

FIGURE 7

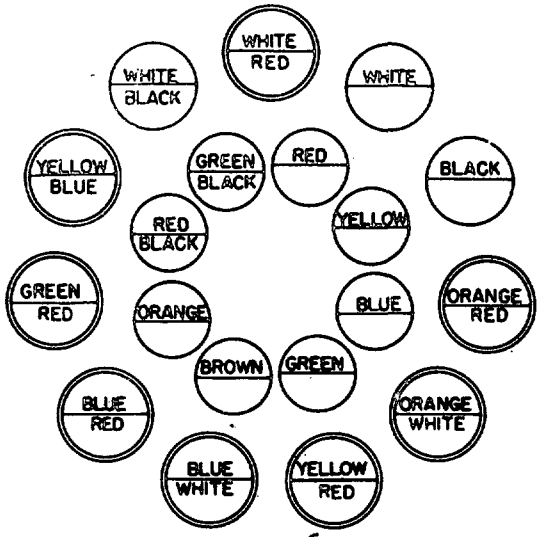
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CABLE TYPES:
S190875 191110S* 191140

*NOTE: DOUBLE CIRCLE INDICATES SHIELDED CONDUCTORS

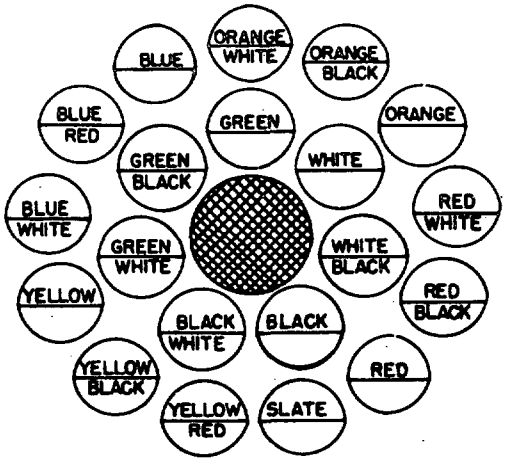
FIGURE 8



CABLE TYPE: S190965S

NOTE: DOUBLE CIRCLE INDICATES SHIELDED CONDUCTORS

FIGURE 9



CABLE TYPE: 200935

FIGURE 10

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